

HT Matrices 2

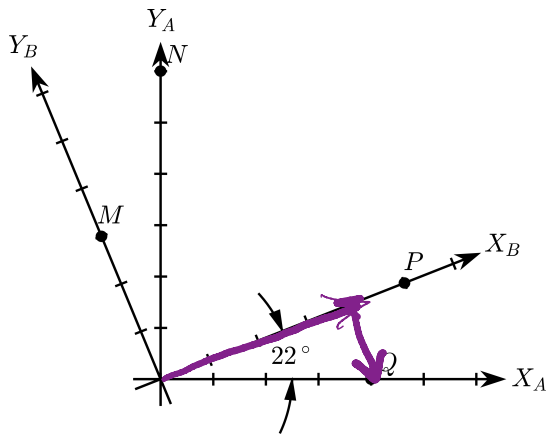
EGR 445/545

GVSU

Three Main Things for Today

- go deeper on HT matrices so you are truly ready for LA 5
- go over two link inverse kinematics so you are ready for project 1
- project time
 - two byte serial
 - controlling servos from Python

Review: LA 3



What do you expect for the signs of the following vectors?:

- Q_B $x+, y-$
- P_A $x+, y+$
- N_B $+$ $+$
- M_A $-$ $+$

Check signs on
Python output

The basis of HT matrices: Eqns. 2.17 - 19

Eqn. 2.17: ${}^A P = \overset{\text{rotation}}{\underbrace{{}^A_B \mathbf{R} \cdot {}^B P}} + \underbrace{{}^A P_{BORG}}_{\text{Translation}}$

Eqn. 2.18: $\begin{bmatrix} {}^A \mathbf{P} \\ 1 \end{bmatrix} = {}^A_B \mathbf{T} \begin{bmatrix} {}^B \mathbf{P} \\ 1 \end{bmatrix}$

Eqn. 2.19: $\begin{bmatrix} {}^A \mathbf{P} \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B \mathbf{R} & {}^A \mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B \mathbf{P} \\ 1 \end{bmatrix}$

same

Goal

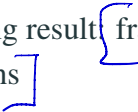
- Our goal is to model any robot using HT matrices like so:

$${}^0P_{tip} = {}^0_1\mathbf{T} \cdot {}^1_2\mathbf{T} \cdot {}^2_3\mathbf{T} \cdot {}^3_4\mathbf{T} \cdot {}^4_5\mathbf{T} \cdot {}^5_6\mathbf{T} \cdot {}^6P_{tip}$$

- the HT matrices will come from eqn. 3.6 and the Denavit-Hartenberg convention
 - we are using JJ Craig's Introduction to Robotics book

Frame attachment and mapping

the big idea:

- we attach frames by starting at the base of the robot and working our way out to the tip
 - we know the tip in the last frame and need to map it back to frame 0
 - possibly confusing result
- 
- frame attachment and mathematical mapping go in opposite directions

Need

- we need to combined rotation and translation into one matrix multiplication operation:

$${}^A P = {}^A \cancel{R} {}^B P + {}^A P_{BORG}$$

Addition takes place in
frame A

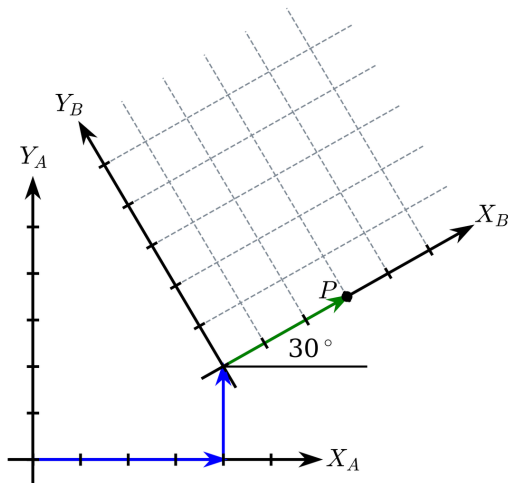
Claim

- a 4x4 HT matrix of this form meets our need for combining rotation and translation:

$${}^A_B\mathbf{T} = \begin{bmatrix} {}^A_B\mathbf{R} & {}^A\mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Adding Vectors

- we cannot add vectors in different frames:



Relationship between frames A and B

Translate First, then Rotate:

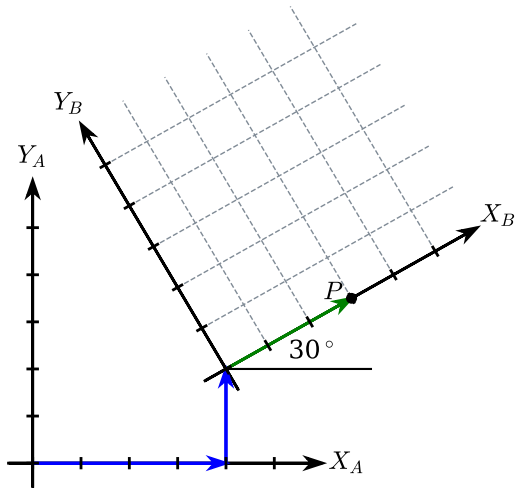
- start with frames A and B coincident
- translate from B by ${}^A P_{BORG}$
- rotate frame B
- note that the mapping is from B back to A

Mathematical Order

- note the order of operations:

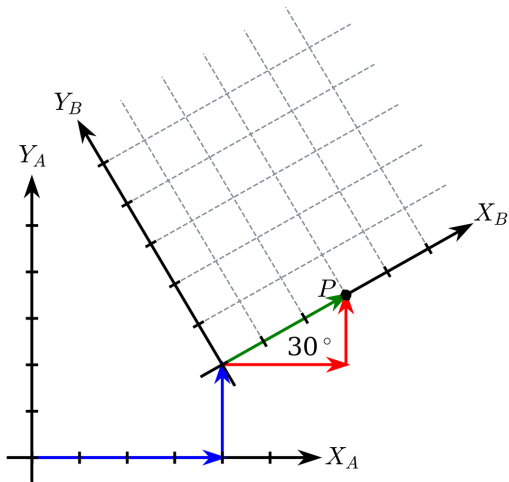
$${}^A P = {}^A_B \mathbf{R} \cdot {}^B P + {}^A P_{BORG}$$

- rotate into frame A
- add the vectors in frame A components



Adding in the Same Frame

- once the vector components are in the same frame, we can add them legally:



HT Matrix Math

$${}^A P = {}^A_B \mathbf{R} \cdot {}^B P + {}^A P_{BORG}$$

becomes

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B \mathbf{R} & {}^A \mathbf{P}_{BORG} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

HT Math R_z Example

$$\begin{matrix}
 & R_z(\theta) & & \text{Translation: } {}^A P_{B_{org}} \\
 & \downarrow & & \swarrow \\
 \begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = & \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 & x \\ \sin(\theta) & \cos(\theta) & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix} & \cdot & \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix}
 \end{matrix}$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P_x \cos(\theta) - {}^B P_y \sin(\theta) + x \\ {}^B P_x \sin(\theta) + {}^B P_y \cos(\theta) + y \\ {}^B P_z + z \\ 1 \end{bmatrix}$$

Translation is added in

Why is the bottom row necessary?

$$\begin{array}{c} 3 \times 4 \end{array} \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 & x \\ \sin(\theta) & \cos(\theta) & 0 & y \\ 0 & 0 & 1 & z \end{bmatrix} \cdot \begin{array}{c} 4 \times 1 \\ \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix} \end{array} = \begin{array}{c} 3 \times 1 \\ \begin{bmatrix} {}^B P_x \cos(\theta) - {}^B P_y \sin(\theta) + x \\ {}^B P_x \sin(\theta) + {}^B P_y \cos(\theta) + y \\ {}^B P_z + z \end{bmatrix} \end{array}$$

- we have the correct answer for ${}^A P$, but the dimensions of the answer are 3×1
- we could not legally multiply by the next 4×4 if we needed to do a repeated mapping:

$${}^0 P_{tip} = {}^0_1 \mathbf{T} \cdot {}^1_2 \mathbf{T} \cdot {}^2_3 \mathbf{T} \cdot {}^3_4 \mathbf{T} \cdot {}^4_5 \mathbf{T} \cdot {}^5_6 \mathbf{T} \cdot {}^6 P_{tip}$$

A General HT Matrix

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & x \\ r_{21} & r_{22} & r_{23} & y \\ r_{31} & r_{32} & r_{33} & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P_x r_{11} + {}^B P_y r_{12} + {}^B P_z r_{13} + x \\ {}^B P_x r_{21} + {}^B P_y r_{22} + {}^B P_z r_{23} + y \\ {}^B P_x r_{31} + {}^B P_y r_{32} + {}^B P_z r_{33} + z \\ 1 \end{bmatrix}$$

What if the bottom row were different?

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} & x \\ r_{21} & r_{22} & r_{23} & y \\ r_{31} & r_{32} & r_{33} & z \\ u & v & w & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P_x r_{11} + {}^B P_y r_{12} + {}^B P_z r_{13} + x \\ {}^B P_x r_{21} + {}^B P_y r_{22} + {}^B P_z r_{23} + y \\ {}^B P_x r_{31} + {}^B P_y r_{32} + {}^B P_z r_{33} + z \\ {}^B P_x u + {}^B P_y v + {}^B P_z w + 1 \end{bmatrix}$$

- the bottom row must be $[0, 0, 0, 1]$ for subsequent multiplications

HT Matrices and my robotics module

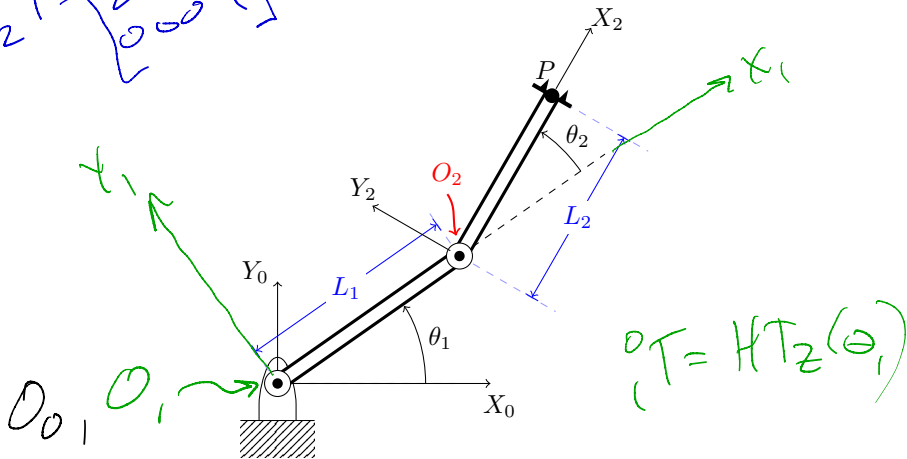
- `robotics.HTz`
 - `robotics.HTx`
 - `robotics.HTy`
- `robotics.HT_from_R_and_Porg`
 - this is foundational
 - what does this code do?

$$\begin{bmatrix} R_y & x \\ & y \\ & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Two-Link Example

$${}^1_2T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$${}^1_2P = R_z(\theta_2)$$



$${}^0_1T = H(T_2(\theta_1))$$

Review: Find ${}^A P$

